16-Bit Microprocessor

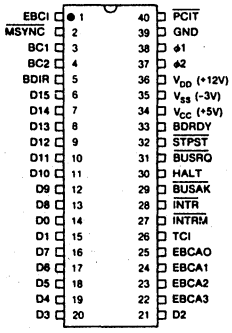
FEATURES

- 8 program accessible 16-bit general purpose registers
- 86 basic instructions
- 4 addressing modes: immediate, direct, indirect, relative
- Conditional branching on status word and 16 external conditions
- Unlimited interrupt nesting and priority resolution
- 16-bit logic and 2's complement arithmetic
- Status logic and word: carry, overflow, sign, zero
- Direct memory access (DMA) for high speed data transfer
- 64K memory using single address
- TTL compatible/simple bus structure
- CP1610: 1μs cycle time, 2MHz 2-phase clock

PIN CONFIGURATION

40 LEAD DUAL IN LINE

Top View



DESCRIPTION

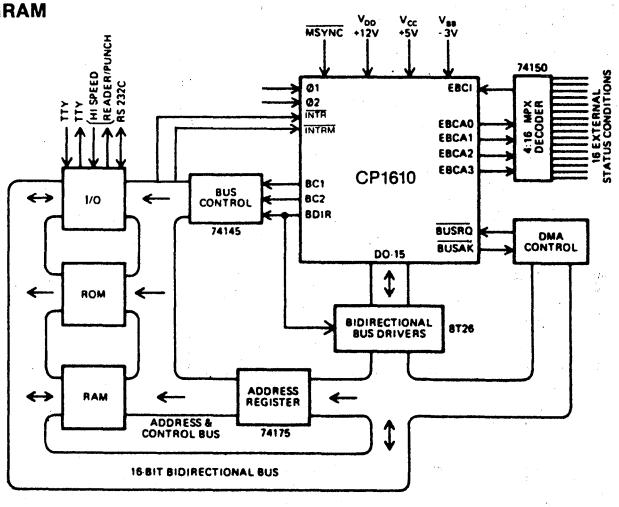
The CP1610 is a compatible member of the Series 1600 Microprocessor products family. It is a complete, 16-bit, single chip, high speed MOS-LSI Microprocessor. The Series 1600 family is fabricated with General Instrument's N-Channel Ion-Implant process, insuring high performance with proven reliability and production history. All members of the Series 1600 family are fully compatible with the CP1610.

The Microprocessor has been designed for high speed data processing and real time applications. Typical applications include programmable TV games, home computer systems/ home information centers, programmable calculator systems, peripheral controllers, process controllers, intelligent terminals and instruments, data acquisition and digital communications processors, numerical control systems and many general purpose mini-computer applications.The Microprocessor can readily support a variety of peripheral equipment such as TTY, CRT display, tape reader/punch, A/D & D/A converter, keyboard, cassette tape, floppy disk, and RS-232C data communication lines.

The CP1610 utilizes third generation minicomputer architecture with eight general purpose registers to achieve a versatile, sophisticated microcomputer system. The 16-bit word enables fast and efficient processing of alphanumeric or byte oriented data. The 16-bit address capability permits access to 65,536 words in any combination of program memory, data memory, or peripheral devices. This single address space concept, combined with a powerful instruction set, provides an efficient solution to microcomputer and many minicomputer-based product requirements.

PERSONAL TERMINAL

CP1610 SYSTEM DIAGRAM



PROCESSOR SIGNALS

DATA BUS

D0-D15
Input/Output/High Impedance

Data 0-15: 16-bit bidirectional bus used to transfer data, addresses, and instructions between the microprocessor, memory, and peripheral devices.

PROCESSOR CONTROL

$\overline{\text{STPST}}$
Input
SToP-STart: Edge-triggered by negative transition; used to control the running condition of the microprocessor.

HALT
Output
HALT: indicates that the microprocessor is in a stopped mode.

$\overline{\text{MSYNC}}$
Input
Master SYNC: Active low input synchronizes the microprocessor to the $\phi 1$, $\phi 2$ clocks during power-up initialization.

EBCA 0-3
Outputs
External Branch Condition Addresses 0-3: Address for one-of-16 external digital state tests via the BEXT (Branch on EXTernal) instruction.

EBCI
Input
External Branch Condition Input: Return signal from the one-of-16 selection made by EBCA 0-3.

BUS CONTROL

BDIR, BC1, BC2
Outputs
Bus DIRection, Bus Controls 1, 2: Bus control signals externally decoded to define the state of bus operations (see State Flow Diagram).

$\overline{\text{BUSRQ}}$
Input

$\overline{\text{BUSAK}}$
Output
BUS ReQuest, BUS AcKnowledge: BUSRQ* requests the microprocessor to relinquish control of the bus indefinitely. BUSAK* informs devices that the bus has been released.

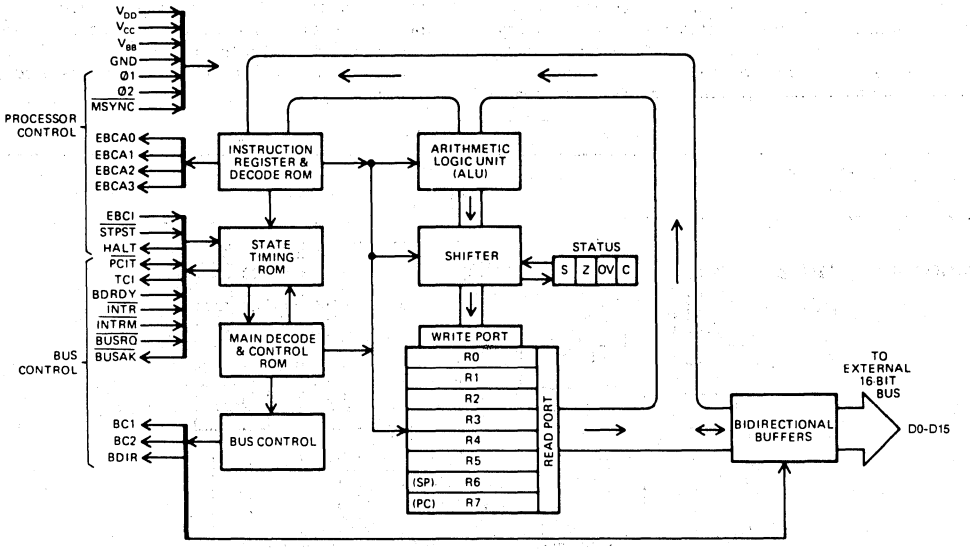
BDRDY
Input
Bus Data ReaDY: causes the microprocessor to "wait" and resynchronize to slow memory and peripheral devices.

$\overline{\text{INTR}}$, $\overline{\text{INTRM}}$
INTeRupt, INTeRupt Masked: request the microprocessor to service an interrupt upon completion of current instruction.

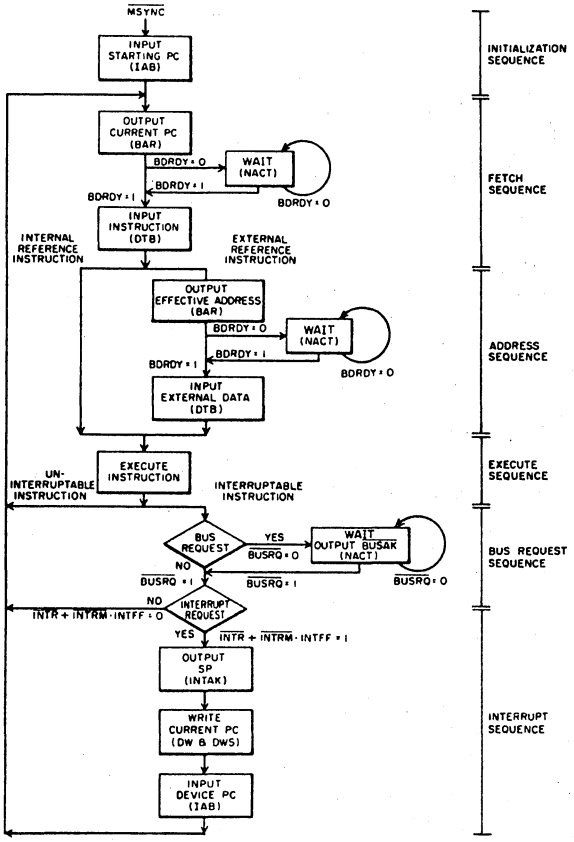
TCI
Output
Terminate Current Interrupt: pulse outputted by the microprocessor in response to the TCI instruction.

$\overline{\text{PCIT}}$
Input/output
Program Counter Inhibit/Trap: As an input, inhibits incrementation of the Program Counter during the instruction fetch sequence. As an output, generates a pulse during execution of a Software INterrupt (SIN) instruction.

CP1610 INTERNAL BLOCK DIAGRAM



SIMPLIFIED STATE FLOW DIAGRAM



BUS CONTROL SIGNALS

BDIR	BC2	BC1	Signal	Decoded Function
0	0	0	NACT	No ACTion, D0-D15 = high impedance
0	0	1	ADAR	Address Data to Address Register, D0-D15 = high impedance
0	1	0	IAB	Interrupt Address to Bus, D0-D15 = Input
0	1	1	DTB	Data To Bus, D0-D15 = Input
1	0	0	BAR	Bus to Address Register
1	0	1	DW	Data Write
1	1	0	DWS	Data Write Strobe
1	1	1	INTAK	INTerrupt AcKnowledge

INSTRUCTION SET (SUMMARY LISTING)

		Mnemonics	Operation	Microcycles				Comments
INTERNAL REFERENCE INSTRUCTIONS	Register to Register	MOVR	MOVe Register	6/7				
		TSTR	TeST Register	6/7				MOVR to itself
		JR	Jump to address in Register	7				MOVR to PC
		ADDR	ADD contents of Registers	6				
		SUBR	SUBtract contents of Registers	6				
		CMPR	CoMPare Registers by subtr.	6				Results not stored
		ANDR	logical AND Registers	6				
		XORR	eXclusive OR Registers	6				
		CLRR	CLeaR Register	6				XORR with itself
	Single Register	INCR	INCrement Register	6				
		DECR	DECrement Register	6				
		COMR	COMplement Register	6				One's Complement
		NEGR	NEGate Register	6				Two's Complement
		ADCR	ADd Carry Bit to Register	6				
		GSWD	Get Status WorD	6				
		NOP	No OPeration	6				
		SIN	Software INterrupt	6				Pulse to $\overline{PCIT}$ pin
		RSWD	Return Status WorD	6				
	Register Shift	SWAP	SWAP 8-bit bytes	6				Not interruptable. One or two position shift capability. Two position SWAP not supported.
		SLL	Shift Logical Left	6				
		RLC	Rotate Left thru Carry	6				
		SLLC	Shift Logical Left thru Carry	6				
		SLR	Shift Logical Right	6				
		SAR	Shift Arithmetic Right	6				
		RRC	Rotate Right thru Carry	6				
		SARC	Shift Arithmetic Right thru Carry	6			Add two cycles for 2-position shift.	
	Control Instructions	HLT	HaLT	4				
		SDBD	Set Double Byte Data	4				Must precede external reference to double byte data
		EIS	Enable Interrupt System	4				
		DIS	Disable Interrupt System	4				Not interruptable
		TCI	Terminate Current Interrupt	4				
		CLRC	CLeaR Carry to zero	4				
		SETC	SET Carry to one	4				
	Jump Instructions	J	Jump	12				
		JE	Jump, Enable, interrupt	12				
		JD	Jump, Disable interrupt	12				
		JSR	Jump, Save Return	12				Return Address saved in R4, 5 or 6.
		JSRE	Jump, Save Return & Enable	12				
		JSRD	Jump, Save Return & Disable interrupt	12				
EXTERNAL REFERENCE INSTRUCTIONS	Conditional Branch Instructions	B	unconditional Branch	9*				Two words
		NOPP	No OPeration	7*				Two words
		BC (BLGE)	Branch on Carry	7				C=1
		BNC (BLLT)	Branch on No Carry	7				C=0
		BOV	Branch on OVerflow	7				OV=1
		BNOV	Branch on No OVerflow	7				OV=0
		BPL	Branch on PLus	7				S=0
		BMI	Branch on Minus	7				S=1
		BZE (BEQ)	Branch on ZEro or EQual	7				Z=1
		BNZE (BNEQ)	Branch if Not ZEro or Not EQual	7				Z=0
		BLT	Branch if Less Than	7				S∀OV=1
		BGE	Branch if Greater than or Equal	7				S∀OV=0
		BLE	Branch if Less than or Equal	7				ZV(S∀OV)=1
		BGT	Branch if Greater Than	7				ZV(S∀OV)=0
		BUSC	Branch if Sign≠Carry	7				C∀S=1
		BESC	Branch if Sign=Carry	7				C∀S=0
		BEXT	Branch if External condition is True	7			Add 2 cycles if test condition is true, except *	4 LSB of instruction are decoded select 1 of 16 external conditions. Displacement in PC+1 PC←PC±Displacement
	I/O			Dir.	Imm.	Indir.	Stack	
		MVO	MoVe Out	11	9	9	9	Not interruptable
		PSHR	PuSH Register to stack	—	—	—	9	PSHR=MVO@R6. Not interruptable
		MVI	MoVe In	10	8	8	11	
		PULR	PULl from stack to Register	—	—	—	11	PULR=MVI@R6.
	Arithmetic & Logic	ADD	ADD	10	8	8	11	
		SUB	SUBtract	10	8	8	11	
		CMP	CoMPare	10	8	8	11	Result not saved
		AND	logical AND	10	8	8	11	
		XOR	eXclusive OR	10	8	8	11	

1 MICROCYCLE=2 CLOCK CYCLES

ELECTRICAL CHARACTERISTICS

Maximum Ratings*

V_{DD}, V_{CC}, GND and all other input/output voltages with respect to V_{BB} . . . . . −0.3V to +18.0V
Storage Temperature . . . . . -55°C to +150°C
Operating Temperature . . . . . .0°C to +70°C

*Exceeding these ranges could cause permanent damage. Functional operation of this device at these conditions is not implied—operating ranges are specified below.

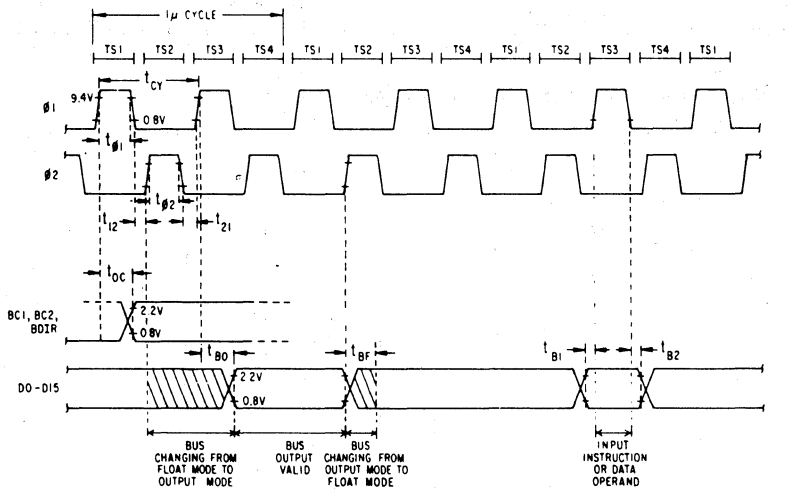
Standard Conditions: (unless otherwise noted)
V_{DD}=+11V±5%, 70mA(typ), 110mA(max.) V_{BB} = −2.2V ± 5%, 0.2mA(typ), 2mA(max.)
V_{CC}=+5V±5%, 12mA(typ), 25mA(max.) Operating Temperature (T_A)=0°C to +70°C

Characteristic	Sym	Min	Typ**	Max	Units	Conditions
DC CHARACTERISTICS						
Clock Inputs						
High	V_{IHC}	10.0	—	V_{DD}	V	
Low	V_{ILC}	0	—	0.6	V	
Input current	I_C	—	—	15	mA	$V_{IHC} = (V_{DD} - 1)V$
Logic Inputs						
Low	V_{IL}	0	—	0.65	V	
High (All Lines except BDRDY)	V_{IH}	2.4	—	V_{CC}	V	
High (Bus Data Ready Line See Note)	V_{IHB}	3.0	—	V_{CC}	V	
Logic Outputs						
High	V_{OH}	2.4	V_{CC}	—	V	I_{OH} = 100μA
Low (Data Bus Lines D0-D15)	V_{OL}	—	—	0.5	V	I_{OL} = 1.6mA
Low (Bus Control Lines, BC1,BC2,BDIR)	V_{OL}	—	—	0.45	V	I_{OL} = 2.0mA
Low (All Others)	V_{OL}	—	—	0.45	V	I_{OL} = 1.6mA
AC CHARACTERISTICS						
Clock Pulse Inputs, φ1 or φ2						
Pulse Width	$t_{\phi 2}$, $t_{\phi 2}$	250		—	ns	
Skew (φ1, φ2 delay)	t_{12}, t_{21}	0	—	—	ns	
Clock Period	t_{cy}	0.5	—	2.0	μs	
Rise & Fall Times	tr, tf	—	—	15	ns	
Master SYNC:						
Delay from φ	tms	—	—	30	ns	
D0-D15 Bus Signals						
Output delay from φ1 (float to output)	t_{BO}	—	—	100	ns	1 TTL Load & 100pF
Output delay from φ2 (output to float)	t_{BF}	—	50	—	ns	
Input setup time before φ1	t_{B1}	0	—	—	ns	
Input hold time after φ1	t_{B2}	10	—	—	ns	
Bus Control Signals BC1,BC2,BDIR						
Output delay from φ1	t_{DC}	—	—	100	ns	
Skew	—	—	—	30	ns	
BUSAK Output delay from φ1	t_{BU}	—	150	—	ns	
TCI Output delay from φ1	t_{TO}	—	200	—	ns	
TCI Pulse Width	t_{TW}	—	300	—	ns	
EBCA output delay from BEXT input	t_{DE}	—	—	150	ns	
EBCA wait time for EBCI input	t_{AI}	—	—	400	ns	
CAPACITANCE						TA = +25°C; V_{DD} = +12V; V_{CC} = +5V; V_{BB} = −3V; tφ1 = tφ2 = 120ns
φ1, φ2 Clock Input capacitance	Cφ1, Cφ2	—	20	30	pF	
D0-D15	—	—	8	15	pF	
All Other	—	—	5	10	pF	

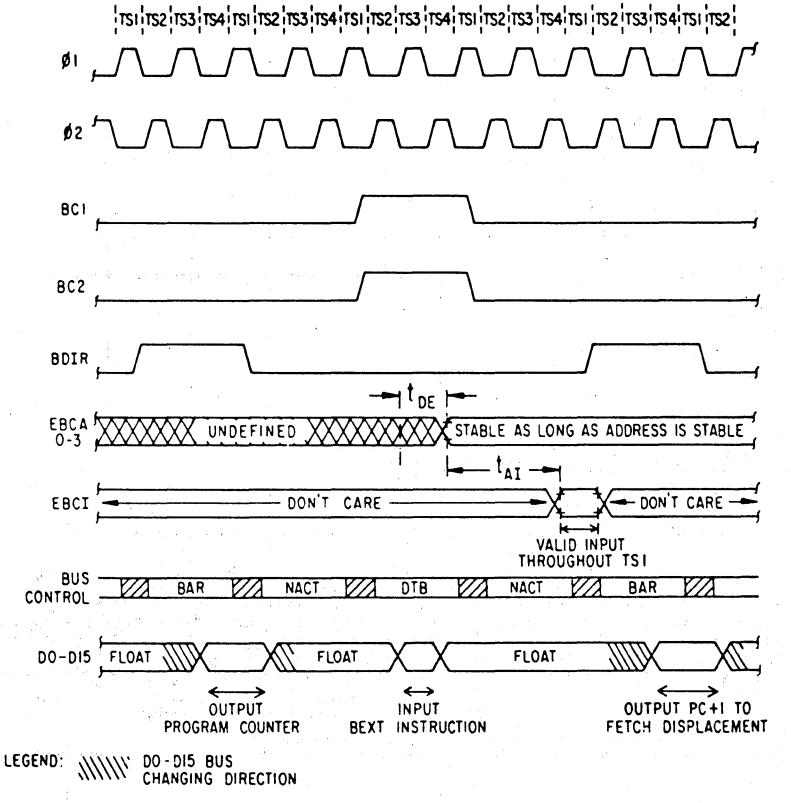
**Typical values are at +25°C and nominal voltages.

NOTE: The Bus Data ReaDY(BDRDY) line is sampled during time period TSI after a BAR or ADAR bus control signal. BDRDY must go low requesting a wait state 50 ns before the end of TS1 and remain low for 50 ns minimum. BDRDY may go high asynchronously. In response to BDRDY, the CPU will extend bus cycles by adding additional microcycles up to a maximum of 40 μsec duration.

CLOCK AND BUS TIMING



TYPICAL INSTRUCTION SEQUENCE (EXTERNAL BRANCH TIMING)



PERSONAL TERMINAL